SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**AcademicYear:**2025-2026
Course Coordinator Name		Venkataramana Veeramsetty	
Instructor(s)Name		1. Dr. Mohammed Ali Shaik 2. Dr. T Sampath Kumar 3. Mr. S Naresh Kumar 4. Dr. V. Rajesh 5. Dr. Brij Kishore 6. Dr Pramoda Patro 7. Dr. Venkataramana 8. Dr. Ravi Chander 9. Dr. Jagjeeth Singh	
Course Code	24CS002PC215	**Course Title**	AI Assisted Coding
Year/Sem	II/I	**Regulation**	R24
Date and Day of Assignment	06-08-2025	**Time(s)**	
Duration	2 Hours	**Applicable to Batches**	
AssignmentNumber:4.5(Present assignment number)/**24**(Total number of assignments)			

Q. No.	Question	*ExpectedTime to complete*
1	**Lab 4: Advanced Prompt Engineering: Zero-shot, one-shot, and few-shot techniques** **Objective:** To explore and compare Zero-shot, One-shot, and Few-shot prompting techniques for classifying emails into predefined categories using a large language model (LLM). Suppose that you work for a company that receives hundreds of customer emails daily. Management wants to automatically classify emails into categories like "Billing", "Technical Support", "Feedback", and "Others" before assigning them to appropriate departments. Instead of training a new model, your task is to use prompt engineering techniques with an existing LLM to handle the classification. Tasks to be completed are as below 1. **Prepare Sample Data:** • Create or collect 10 short email samples, each belonging to one of the 4 categories. **1. Billing** – "I need a breakdown of my last payment for accounting purposes." – "Can you confirm if my subscription was renewed this month?" – "Please update my billing address to the new location." **2. Technical Support** – "The software keeps freezing when I try to export a file."	08.08.2025 EOD

– "I'm unable to log in even after resetting my password."
– "Can someone help me fix this installation error?"

3. Product Inquiry

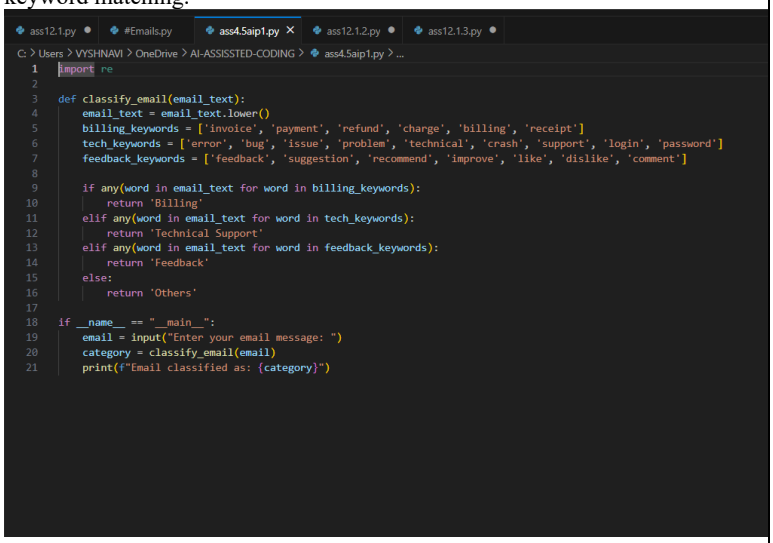
– "Is the black model of this phone available in 128GB?"
– "Does this plan include international calling?"
– "Can you tell me if this product is compatible with Windows 11?"

4. General Feedback

– "Your team was incredibly helpful during my last visit—thank you!"
– "I love how intuitive your new dashboard design is."

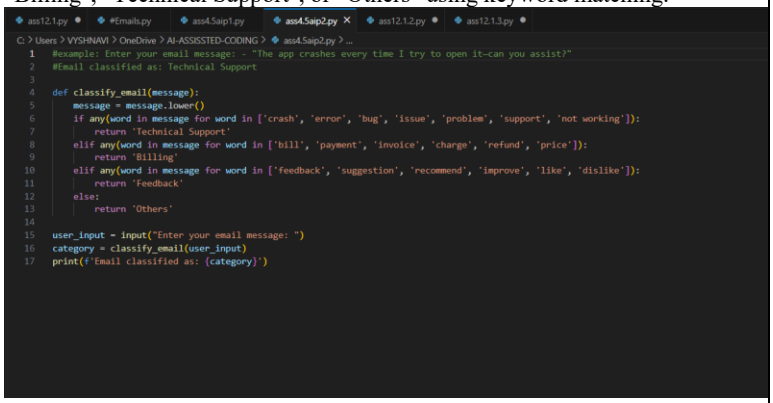
2. **Zero-shot Prompting:**

- Design a prompt that asks the LLM to classify a single email without providing any examples.
- Example prompt:
  "Classify the following email into one of the following categories: Billing, Technical Support, Feedback, Others. Email: 'I have not received my invoice for last month.'"
  PROMPT:
  Classify user-entered email messages into categories using Python and keyword matching.



3. One-shot Prompting:

- Add one labeled example before asking the model to classify a new email.
  PROMPT:
  Write a Python program that asks the user for a message and classifies it as "Billing", "Technical Support", or "Others" using keyword matching.



4. **Few-shot Prompting:**

- Use 3–5 labeled examples in your prompt before asking the model to classify a

new email.

PROMPT:

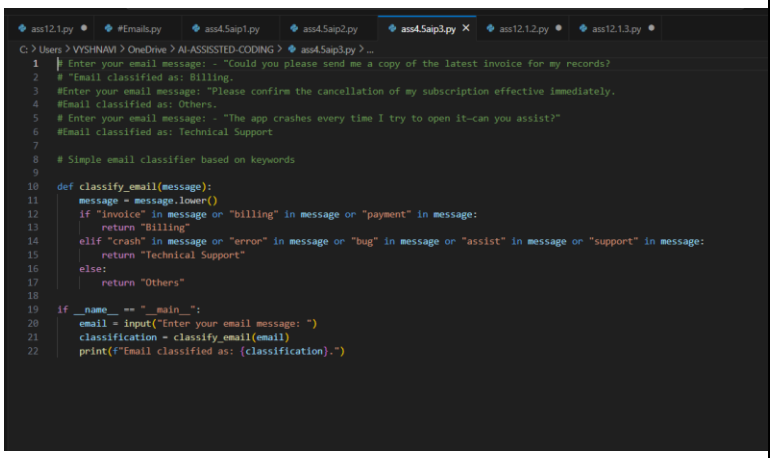
Write a Python program that categorizes customer feedback based on its content.
The program should:

– Ask the user to enter a feedback message

– Use keyword matching to classify the message into one of three categories:

- "Product Inquiry" (e.g., contains words like "price", "availability", "details")
- "Complaint" (e.g., contains words like "broken", "late", "bad", "issue")
- "General Feedback" (if no keywords match)

– Print the classification resul



5. **Evaluation:**

- Run all three techniques on the same set of 5 test emails.
- Compare and document the accuracy and clarity of responses.

OBSERVATION:

The code is **accurate**, **clear**, and **well-aligned** with the prompt. It uses straightforward logic and clean structure to classify emails effectively. Ideal for basic automation or as a starting point for more advanced NLP tasks.

Requirements:

- VS Code with Github Copilot or Cursor IDE and/or Google Colab with Gemini

Deliverables:

- A .txt or .md file showing prompts and model responses.
- A comparison table showing classification accuracy for each technique.
- A short reflection on which method was most effective and why

.